

A Low-Cost Hardware-in-the-Loop Testbed for Distribution Grid Volt-Var Control: Architecture, Benchmarking, and Quantisation Effects

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Abstract—Rising penetration of distributed energy resources (DERs) increasingly causes overvoltage violations in medium-voltage (MV) distribution grids. This paper presents a low-cost Hardware-in-the-Loop (HIL) testbed in which a Raspberry Pi 5 executes pandapower/SimBench MV networks while an Arduino Uno R3 implements the VDE-AR-N 4110 Q(V) characteristic as a physical reactive power controller, augmented by a Jacobian-based sensitivity coordinator. On a rural SimBench MV feeder, sensitivity coordination reduces active power curtailment by up to 46.2 % relative to local Q(V)-only control. Beyond benchmarking, we report a novel quantisation finding: the four-decimal ASCII serial protocol perturbs reactive power setpoints just enough to raise the coordinator activation rate from 9.5 % (pure-software dry-run) to 99.9 % (HIL) on the same network—directly analogous to Modbus/SunSpec 16-bit register truncation in field inverters. Idealised simulation therefore underestimates real-world coordinator activation, with implications for field deployment of sensitivity-based Volt-Var control.

Index Terms—Hardware-in-the-loop, Volt-Var control, reactive power, distribution grids, quantisation, SimBench, pandapower, VDE-AR-N 4110

I. INTRODUCTION

The rapid growth of distributed energy resources (DERs), particularly rooftop and utility-scale photovoltaics, is transforming medium-voltage (MV) distribution grids from passive load-serving networks into active systems in which reverse power flows routinely push bus voltages above their statutory limits. Reactive power provision by DER inverters is a well-established and economically attractive lever for voltage regulation under high penetration, as it exploits latent inverter capability without sacrificing energy yield [1]. Where reactive power alone is insufficient, active power curtailment becomes the remedy of last resort, directly reducing renewable energy harvest [2]. In Germany, the grid connection standard VDE-AR-N 4110 formalises this hierarchy for MV-connected plants by mandating a piecewise-linear Q(V) characteristic (Bild 8) executed locally by each generating unit [3].

Validating voltage control strategies before field deployment requires test environments that reproduce not only the network physics but also the behaviour of the real control hardware. Hardware-in-the-Loop (HIL) and power-HIL platforms are the established answer [4], [5], and have been applied to distribution-scale problems including remote circuit co-simulation [6]. However, commercial real-time simulators (RTDS, OPAL-RT and similar) carry costs that place them out

of reach for many teaching and early-stage research settings; low-cost alternatives exist [7] but focus on power-stage emulation rather than the communication and arithmetic precision of the controller itself. Conversely, pure software simulation represents DER controllers with float64 arithmetic and loss-less information exchange—a fidelity that no fielded inverter possesses, since real devices communicate setpoints through finite-precision channels such as 16-bit Modbus/SunSpec registers [8], [9], [10]. Neither extreme exposes how controller-side quantisation propagates into the behaviour of higher-level coordination layers.

This paper contributes: (i) a low-cost HIL testbed (Raspberry Pi 5 plus Arduino Uno R3) in which a real embedded device executes the VDE-AR-N 4110 Q(V) characteristic inside a pandapower/SimBench time-series simulation; (ii) a reproducible five-scenario benchmark framework comparing no control, on-load tap changer (OLTC), static VAr compensator (SVC), local Volt-Var, and Jacobian-based coordinated Volt-Var at full annual resolution, with an extensible plugin architecture for third-party controllers and networks; (iii) a timing characterisation of the HIL loop demonstrating real-time feasibility; and (iv) a quantisation finding with direct field-deployment implications: hardware-induced truncation of voltage telemetry raises the activation rate of a sensitivity-based coordinator from 9.5 % to 99.9 % on the same network. The remainder of the paper describes the testbed architecture (Section II), the benchmark framework (Section III), results including the quantisation effect (Section IV), discussion (Section V), and conclusions (Section VI).

II. TESTBED ARCHITECTURE

A. Hardware

The testbed comprises two commodity devices. A Raspberry Pi 5 (8 GB, EUR 189.44) acts as the grid simulator, running the full Python power system stack. An Arduino Uno R3 is physically inserted into the control loop as the reactive power controller, connected over USB serial at 115 200 baud (8N1). The Arduino firmware implements the VDE-AR-N 4110 Q(V) Bild 8 piecewise-linear characteristic with breakpoints $U_1=0.96$, $U_2=0.99$, $U_3=1.01$, and $U_4=1.04$ pu and a reactive capability ratio $Q_{\max}/P_{\text{inst}} = 0.25$ per DER: reactive power is injected below U_2 , zero in the deadband $[U_2, U_3]$, and absorbed

Figure placeholder:

Block diagram of the HIL loop (manual drawing; see comment in source).

Fig. 1. HIL loop architecture and per-timestep sequence. Each 15 min simulation step executes: (1) write profile active power to the network model; (2) pre-control power flow yielding bus voltages at all DERs; (3) serial exchange—the RPi transmits per-DER voltage magnitudes (V : message) and the Arduino returns $Q(V)$ setpoints (Q : message), both as four-decimal ASCII; (4) optional Jacobian-based sensitivity coordination of the returned setpoints; (5) PT1 dynamics and ramp limiting; (6) post-control power flow and violation detection; (7) iterative active power curtailment if violations persist.

on a linear ramp between U_3 and U_4 , saturating at $\pm Q_{\max}$ outside the band.

The firmware targets the ATmega328P with hard embedded constraints: of the 2048 B SRAM, 1918 B are allocated (approximately 130 B of stack, roughly 50 B free margin), supporting up to 105 DERs per exchange. All floating-point arithmetic is 32-bit (AVR `double` $\equiv$ `float`). These constraints are deliberately representative of resource-constrained embedded controllers in real DER applications and are, as Section IV-C shows, scientifically productive rather than incidental.

B. Software Stack

The simulation side is built on pandapower 3.4.0 [11] for Newton–Raphson power flow and network modelling, and SimBench 1.6.1 [12] for benchmark MV networks with native annual profiles. Weather-driven profile generation for non-SimBench networks uses pvlib [13] with German Weather Service (DWD) station data.

Each Scenario 4 timestep performs two power flow solutions: a pre-control `runpp()` to obtain the voltage magnitudes sent to the Arduino, and a post-control `runpp()` after reactive power and dynamics are applied, followed by violation detection and, if needed, an iterative curtailment loop (10% reduction per iteration, at most 10 iterations). The `SensitivityCoordinator` implements the coordinated path: it linearises the voltage–reactive-power relationship through the power flow Jacobian [14] and, whenever the predicted post- $Q(V)$ voltage still exceeds the violation threshold at any bus, solves a least-squares problem over the sensitivity submatrix of violated buses for a minimum-norm reactive power correction distributed across all DERs—a sensitivity-based Volt-Var philosophy shared with discrete coordinate-descent approaches [15]. On rural radial feeders this submatrix is structurally rank-deficient (rank 3–4 despite 8+ DERs, since DERs on the same branch produce collinear sensitivity

TABLE I
BENCHMARK SCENARIOS

No.	Name	Control method	HW
1	Baseline	None (violation reference)	N
2	OLTC	On-load tap changer	N
3	SVC	Static VAR compensator at MV busbar	N
4A	Volt-Var local	Arduino $Q(V)$, VDE-AR-N 4110 Bild 8	Y
4B	Volt-Var coord.	Arduino $Q(V)$ + Jacobian sensitivity coordinator (least-squares correction)	Y
5	OPF benchmark	AC optimal power flow (bound)	N

columns); the SVD-based least-squares solve handles the null space directly. Applied setpoints are finally shaped by a first-order (PT1) reactive power filter with active power ramp limiting (`DERDynamics`), emulating inverter response dynamics.

C. Extensibility: Controller and Network Plugins

Because the testbed serves as a shared research platform, both the controller and the network layer are extensible without modification of any core module. A controller plugin system loads user-defined $Q(V)$ (or arbitrary Q) control functions from YAML descriptors, validated against a safety matrix (identifier collisions, hardware guards, capability clamping through a dry-run controller instance) and registered dynamically alongside the built-in scenarios; reference plugins include a proportional Q – P droop and a four-breakpoint deadband characteristic using VDE-AR-N 4110 default parameters. A companion network plugin system admits external grids from pandapower JSON, pickle, or loader functions, with three profile strategies (SimBench-native, DWD/pvlib weather-driven, or flat rated-capacity), so that student-contributed controllers and networks run under the identical benchmark, metric, and publication pipeline as the built-in scenarios.

III. FIVE-SCENARIO BENCHMARK FRAMEWORK

Table I summarises the benchmark scenarios. Scenarios 1–3 and 5 are pure software; only Scenario 4 (both variants) places the Arduino in the loop, and each Scenario 4 variant can additionally run in a *dry-run* mode in which a float64 Python implementation of the identical $Q(V)$ characteristic replaces the Arduino—the comparison that enables the quantisation analysis of Section IV-C. Scenario 5 provides a theoretical optimum bound via AC optimal power flow and runs on the CIGRE MV network only, because the HV/MV transformer impedance contrast of SimBench MV networks ill-conditions the OPF admittance matrix.

The primary benchmark network is SimBench 1-MV-rural--2-sw (99 buses, 102 DERs including 91 aggregated low-voltage residential PV units, closed sectionalizer). All annual runs use the native SimBench profiles at 15 min resolution: 35 136 timesteps covering the full 366-day leap year 2016. The open-sectionalizer variant 1-MV-rural--1-sw serves as the secondary network on which the primary coordination result is demonstrated. Reported metrics are: the number of violated timesteps ($V_{\min}=0.95$, $V_{\max}=1.05$ pu); the voltage deviation

TABLE II
ANNUAL BENCHMARK, SIMBENCH 1-MV-rural--1-sw (CLOSED
SECTIONALIZER), 35 136 TIMESTEPS, $Q_{\max}/P_{\text{inst}} = 0.25$

Scenario	Viol. steps	max v_m (pu)	Curtail. steps	Coord. rate
1 Baseline	2880	[TBD]	—	—
2 OLTC	862	[TBD]	—	—
3 SVC ^a	[TBD]	[TBD]	—	—
4A Local	955 ^b	[TBD]	290	0 % ^c
4B Coord.	490 ^b	[TBD]	156	9.1 %

^aSVC fails catastrophically on this topology (min $v_m = 0.827$ pu) and is reported as a negative result. ^bOvervoltage timesteps; VDI = 3960 (4A) vs. 3879 (4B). ^cZero by definition: no coordination layer is active in 4A.

index $\text{VDI} = \sum_t \sum_b |v_{b,t} - 1.0|$ summed over all buses and converged timesteps (dimensionless, additive across subnetworks); the maximum bus voltage max v_m ; the number of timesteps requiring active power curtailment; and the coordination rate, the share of timesteps on which the sensitivity coordinator modifies the local Q(V) setpoints.

The Q(V) characteristic follows VDE-AR-N 4110 [3] rather than IEEE 1547-2018 [10] because the testbed models German MV networks, for which VDE-AR-N 4110 is the governing connection standard; its Bild 8 characteristic prescribes the breakpoint voltages and deadband adopted by the firmware. IEEE 1547-2018 Category B volt-var is functionally analogous but differs in default breakpoints, so results transfer qualitatively rather than numerically; the firmware curve is parameterised and can be re-pointed to IEEE 1547 defaults without structural change.

IV. RESULTS

A. Voltage Control Performance

Table II reports the annual benchmark on 1-MV-rural--1-sw (closed sectionalizer). The uncontrolled baseline violates limits on 2880 timesteps; the OLTC reduces this to 862. The SVC is included honestly as a negative result: on this radial rural topology it is catastrophic, collapsing the minimum bus voltage to 0.827 pu, and is therefore unsuitable for this network class. The two Volt-Var variants form the main comparison: local Q(V) control (Scenario 4A) requires curtailment on 290 timesteps, whereas adding the sensitivity coordinator (Scenario 4B) reduces this to 156 timesteps—a 46.2 % reduction in curtailment events—while also reducing overvoltage timesteps from 955 to 490 and the VDI from 3960 to 3879, at a coordination rate of only 9.1 %. The coordinator thus intervenes sparingly but at the timesteps where local control alone would otherwise force curtailment. Consistent behaviour is observed on the other network configurations: −45.9% curtailment on the open-sectionalizer variant of the same network and −31.6% on 1-MV-rural--2-sw (open sectionalizer).

B. HIL Timing Characterisation

The Arduino serial round trip (hil_latency_ms) has a mean of 151.8 ms with a standard deviation below 1.2 ms—a

Figure placeholder:
Annual voltage envelope (min/max bus voltage per timestep) for
Scenarios 1, 2, 4A and 4B; from compare_scenarios.py.

Fig. 2. Annual voltage envelope comparison on 1-MV-rural--1-sw. The coordinated Volt-Var scenario (4B) suppresses the overvoltage excursions that force curtailment under local control (4A), while the baseline exhibits sustained violations of the 1.05 pu limit.

tight, unimodal distribution with no drift over the full annual run. The median total per-timestep cost on the Raspberry Pi is approximately 227 ms for Scenario 4A and 234 ms for Scenario 4B, so the serial exchange accounts for 66 % (4A) and 64 % (4B) of the timestep budget; the non-serial residual of approximately 75 ms to 83 ms covers the two power flow solutions, the sensitivity solve, and the DER dynamics. Since one simulated timestep represents 15 min of grid time, the loop operates with real-time margin exceeding three orders of magnitude. Comparing platforms, the Raspberry Pi is *faster* than the reference laptop at pure pandapower computation (ratio $0.62\times$ on Scenarios 1–3) but $1.76\text{--}1.86\times$ slower on Scenario 4, entirely attributable to the serial exchange.

A reproducible outlier pattern was identified in the OLTC scenario: the five largest-latency timesteps occur at identical timestep indices across two independent Raspberry Pi runs, spaced roughly 7000 timesteps (≈ 7 weeks) apart, with monotonically increasing magnitude (1340 ms to 1952 ms). No tap operation was attempted on any of these steps; the signature is attributed to pandapower sparse matrix cache growth over the annual run, and a similar periodicity appears in the SVC scenario.

C. Hardware Quantisation Effect

The central novel finding concerns how finite hardware precision alters coordinator behaviour. The serial protocol transmits each voltage magnitude as four-decimal ASCII, quantising v_m to steps of $\Delta v_m = 10^{-4}$ pu. On the absorption ramp of the Bild 8 characteristic, the corresponding reactive power sensitivity per DER is

$$\Delta q = \frac{Q_{\max}}{U_4 - U_3} \Delta v_m, \quad (1)$$

which for the largest unit in the network ($P_{\text{inst}} = 4.9$ MW, hence $Q_{\max} = 0.25 \times 4.9 = 1.225$ Mvar) evaluates to $1.225 \times 10^{-4} / 0.03 \approx 0.004$ Mvar per quantisation step, and to approximately 0.0013 Mvar per DER on average across the fleet. Aggregated over the 98 profiled DERs, the HIL path thus returns a total reactive power response smaller by roughly

Figure placeholder:

Coordinator activation: dry-run (float64) vs. HIL (4-decimal ASCII) per-day activation counts on the same network; from the publisher JSON of both annual Scenario 4B runs.

Fig. 3. Effect of hardware quantisation on coordinator activation. The four-decimal ASCII truncation of voltage telemetry shifts the Jacobian-predicted voltage marginally above the 1.051 pu gate on nearly all violated timesteps, raising the annual activation rate from 9.5 % (dry-run) to 99.9 % (HIL).

0.13 Mvar than the float64 dry-run path at the same operating point.

This seemingly negligible deficit is decisive at the coordinator gate. The coordinator fires when the Jacobian-predicted post-Q(V) voltage exceeds the violation threshold of $1.05 + 0.001 = 1.051$ pu (band limit plus a 1 mpu noise guard) at any bus. Debug instrumentation of the coordination step confirms the mechanism directly: on a peak summer day (day 172, pre-control maximum 1.0787 pu), the float64 dry-run setpoints pull the predicted voltage below 1.051 pu on *every* timestep—the residual violation mask is empty all day—whereas on the Raspberry Pi the identical timesteps show populated residual masks and active coordination. Aggregated over the year, the coordinator is active on 99.9 % of violated timesteps in HIL mode versus 9.5 % in the dry-run on the same network. The dry-run figure is not idle behaviour: on a more demanding day (day 160, pre-control maximum 1.0888 pu) the dry-run coordinator also fires on 15 afternoon timesteps with residual masks of 1–4 buses and corrections up to 0.095 Mvar, reducing curtailment from 25.3 to 20.7 MW-steps (−18.4%)—confirming genuine coordination work on hard days even under float64 precision.

The correct interpretation is therefore not that either mode is wrong, but that the dry-run establishes the *theoretical minimum* coordinator activation while HIL exposes the *realistic* rate under hardware quantisation; the gap measures the sensitivity of the coordination gate to field device precision. The mechanism is directly analogous to real inverter communications: SunSpec Modbus, the default DER interoperability profile referenced by IEEE 1547-2018 [10], encodes measurements and setpoints as 16-bit integer registers with power-of-ten scale factors [8], [9], imposing quantisation of exactly the kind the ASCII protocol reproduces. A coordination layer calibrated exclusively against idealised float64 simulation will consequently underestimate its own field activation frequency—a gap that pure simulation cannot capture and that this low-cost testbed exposes by construction.

V. DISCUSSION

The quantisation finding has a direct consequence for field deployment of sensitivity-based coordinators. Activation frequency drives communication load, actuator wear-equivalent behaviour (setpoint churn), and the operating envelope within which the coordinator’s linearised model must remain valid; a design tuned against idealised float64 simulation will meet a materially different duty cycle in the field, where every measurement and setpoint passes through finite-precision registers. Coordinator gates—violation thresholds, noise guards, and hysteresis—should therefore be calibrated against the realistic precision of the deployment communication stack (e.g., 16-bit SunSpec register resolution), and HIL evaluation with representative quantisation should precede parameter freeze. The testbed presented here provides exactly this capability at commodity cost, since the precision-limiting element is a real embedded device rather than a simulated abstraction.

Several limitations bound the claims. First, the simulation is quasi-static: successive power flow snapshots at 15 min resolution capture voltage regulation behaviour but not electromagnetic or electromechanical dynamics, converter control interactions, or sub-minute variability. Second, a single Arduino computes the Q(V) response for all DERs in one batched exchange; while this faithfully reproduces embedded arithmetic and protocol quantisation, it does not capture per-inverter heterogeneity, asynchronous communication, or packet loss that a one-device-per-DER architecture would exhibit. Third, the profiles are SimBench synthetic time series rather than measured field data, so absolute violation and curtailment counts are benchmark-relative quantities; the relative comparisons between scenarios, and the quantisation mechanism itself, are the transferable results.

VI. CONCLUSION

This paper presented a low-cost HIL testbed in which a Raspberry Pi 5 runs full-year pandapower/SimBench MV simulations while an Arduino Uno R3 physically executes the VDE-AR-N 4110 Q(V) characteristic, wrapped in a reproducible five-scenario benchmark with an extensible plugin architecture. On a rural SimBench MV feeder, Jacobian-based sensitivity coordination reduced active power curtailment events by 46.2 % relative to local Q(V) control, at a mean HIL serial latency of 151.8 ms. The principal finding is that four-decimal telemetry quantisation—analogueous to 16-bit Modbus/SunSpec register encoding in field inverters—raises the coordinator activation rate from 9.5 % to 99.9 % on the same network, demonstrating that idealised simulation systematically underestimates real-world coordinator duty. Future work extends the benchmark to low-voltage SimBench networks, replaces the ASCII protocol with a real inverter Modbus/SunSpec interface, and couples the testbed to dynamic simulation.

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